# FIITJEE

# ALL INDIA TEST SERIES

## FULL TEST – III

## JEE (Advanced)-2021

### PAPER –2

### TEST DATE: 18-01-2021

# ANSWERS, HINTS & SOLUTIONS

## *Physics*

### PART – I

#### SECTION – A

1. AD

Sol. $\omega_A = \frac{3v}{2R}, \omega_C = \frac{v}{2R}$ $v_A = \frac{v}{2}(\rightarrow), ; v_C = v/2(\leftarrow)$

$k_A = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{CM}.\omega^2 = \frac{11}{16}mv^2$

$k_B = \frac{3}{16}mv^2$

2. ACD

Sol. $E_x = \frac{k}{R^2}\int_0^{2\pi}(\lambda_0 \cos\theta)Rd\theta.\cos\theta = \frac{k\lambda_0}{R} \times \pi$

3. ABC

Sol. Basic concept.

4. ABC

Sol. Impulse of normal reaction on ball $J_y = 2mv\cos\theta$

Impulse of friction $J_x = \mu J_y = 2\mu mv\cos\theta$

In x – direction $mv'\sin\phi - mv\sin\theta = \pm 2\mu mv\cos\theta$

In y – direction $mv'\cos\phi = mv\cos\theta$

Solving we get $\tan\phi - \tan\theta = \pm\ 2\mu$

$J_x.R = I.sw$

5. AC

Sol. $\vec{E} = \dfrac{K.q\left(\vec{r_2} - \vec{r_1}\right)}{|\vec{r_2} - \vec{r_1}|^3}$

$V = \dfrac{R.q}{|\vec{r_2} - \vec{r_1}|}$

6. A

Sol. $Td\theta = dm\omega^2 R$

$T = \lambda\omega^2 R^2$

$KE = \dfrac{1}{2}(\lambda 2\pi R)(wR)^2 = \pi\lambda R^3\omega^2$

$PE = \dfrac{(Stress)^2}{2Y} \times volume = \dfrac{\pi\lambda^2\omega^4 R^5}{Y.a}$

## SECTION – B

7. 9

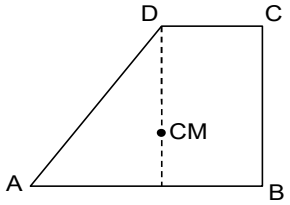
Sol. $I_A = \dfrac{V_0}{\sqrt{R^2 + x_C^2}} = \tan\phi = \dfrac{x_C}{R} = \dfrac{1}{WCR}$

$X_C = \dfrac{1}{WC} = \dfrac{1}{100 \times 10^{-4}} = 100 \tan\phi = \dfrac{100}{50} = 2$

$V_A - V_B = \dfrac{V_0}{Z}.R.\sin\left(100t + \tan^{-1}2\right) - \dfrac{V_0}{2}\sin(100\ t)$

8. 1

Sol. Centre of mass of system lies in vertical line through D.




9. 8

Sol. $\phi = \int B.2\pi r dr$

$e = \dfrac{d\phi}{dt} = E.2\pi r$

$\tau = \int E.\dfrac{Q}{\pi R^2}.2\pi r dr.r$

$\alpha = \dfrac{\tau}{I}$

10. 1

Sol. $\phi = \dfrac{q}{\epsilon_0}(\cos\theta_1 \sim \cos\theta_2)$

$\cos\theta_1 = \dfrac{3}{5}, \cos\theta_2 \dfrac{4}{5}$

11. 6

Sol. If. V is speed at maximum height ‘h’.

$$mv(R+h) = m\sqrt{\frac{7GM}{4R}} \times R$$

$$-\frac{GMm}{R+h} + \frac{1}{2}mv^2 = -\frac{GMm}{R} + \frac{1}{2}m\left(\frac{7GM}{4R}\right)$$

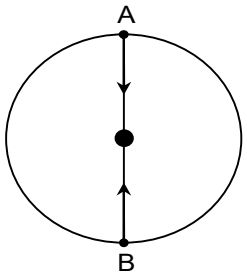
12. 1

Sol. $\frac{4T}{R} = \frac{\sigma^2}{2\epsilon_0} \Rightarrow T = \frac{1\times10^{-12}}{8\epsilon_0}$

## SECTION – C

13. 00320.00

Sol. $a_{rel} = 2\omega^2 R$




$$= 2v^2 / R = 2 \times \frac{16}{(1/5)} = 320 \text{ m/s}^2$$

14. 00003.50

Sol. $R = \int_0^L \frac{dx}{\sigma A} = \frac{L}{\sigma_0 A}\ell n2$

$$j = \frac{V_0}{A.R} = \frac{V_0\sigma_0}{L.\ell n2}$$

$$E = \frac{j}{\sigma} = \frac{V_0}{L(1+x/L)};$$

At mid point $E = \frac{V_0}{1.5\ L.\ \ell n\ 2}$

15. 02160.00

Sol. $\tau = Iw_1w_2$

16. 00007.84

Sol. Basic Concept.

17. 00020.00

Sol. $\rho.\frac{4}{3}\pi r^3 \times 500 \times 800 = \pi r^2 h\ \rho_{ice} \times 80 \times 4.2 \times 10^3$

18. 00004.00

Sol. $T_A + T_C + 2T_B = Mg$

$T_A.\ell/2 = 2T_B.\ell/2$

$T_A = 4Mg/11 \Rightarrow n = 4$

# Chemistry

## PART – II

### SECTION – A

19. ABCD

Sol. $Na_2C_2O_4 \xrightarrow[H^+]{MnO_4^-} CO_2$

$$Na_2C_2O_4 \xrightarrow[\text{hydrolysis}]{\text{Acid}} \begin{matrix}COOH\\|\\COOH\end{matrix} \xrightarrow[\Delta]{H_2SO_4} CO_2 + CO$$

$$\begin{matrix}COOH\\|\\COOH\end{matrix} \xrightarrow{110^0\ C} \begin{matrix}OH\\|\\OH\\|\\OH\end{matrix} \quad H-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[\Delta]{H_2SO_4} CO$$

20. ABCD

Sol. Theory based.

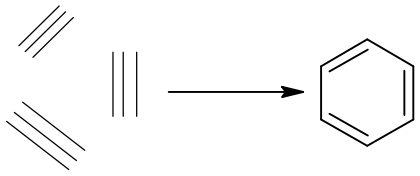
21. AB

Sol. Cisplatin is used for cancer treatment.
Dinitrogen trioxide is a blue solid.

22. D

Sol. (A) $KE = \frac{7.5}{30} \times \frac{3}{2} \times RT = \frac{3}{8}RT$

(B)



$$V = \frac{nRT}{P} = \frac{3 \times 0.1\,RT}{P} = \frac{3}{10} \times \frac{RT}{P}$$

(D) $\frac{r_{H_2}}{r_{NH_3}} = \sqrt{\frac{M_{w_{NH_3}}}{M_{w_{H_2}}}}$

$= \sqrt{\frac{17}{2}} = 2.92$

23. ABCD

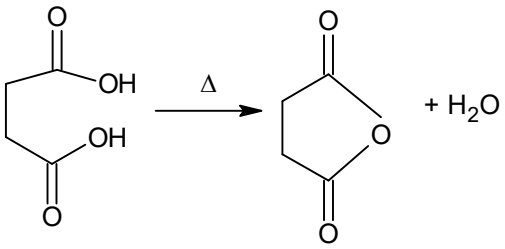
Sol. Theory based

24. AD

Sol.

$$H-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[H_2SO_4]{\text{conc}} CO + H_2O$$

Decarbonylation




## SECTION – B

25. 5

Sol. $M_{(Na_2CO_3)} = \frac{0.6625}{106} \times \frac{1}{0.25} = 2.5 \times 10^{-2} M$

$M_{(KOH)} = \frac{2.8}{56} \times \frac{1}{0.25} = 2 \times 10^{-1} M$

With methyl orange,

$2 \times 2.5 \times 10^{-2} \times 20 \times 10^{-3} + 1 \times 2 \times 10^{-1} \times 20 \times 10^{-3} = \frac{1}{10} \times V$

$10^{-3} + 4 \times 10^{-3} = \frac{1}{10} V$

$5 \times 10^{-2} = V(L)$

$V(mL) = 50\ mL$

With phenolphthalein indicator,

$1 \times 2.5 \times 10^{-2} \times 20 \times 10^{-3} + 1 \times 2 \times 10^{-1} \times 20 \times 10^{-3} = \frac{1}{10} \times V$

$0.5 \times 10^{-3} + 4 \times 10^{-3} = \frac{1}{10} V$

$4.5 \times 10^{-2} = V(L)$

$V(mL) = 45\ mL$

$x - y = 50 - 45 = 5$

26. 1

Sol. $hv = hv_0 + KE$

$KE = h(v - v_0)$

$\frac{KE}{4KE} = \frac{2 \times 10^{-15} - v_0}{5 \times 10^{-15} - v_0}$

$8 \times 10^{-15} - 4v_0 = 5 \times 10^{-15} - v_0$

$+3v_0 = +3 \times 10^{-15}$

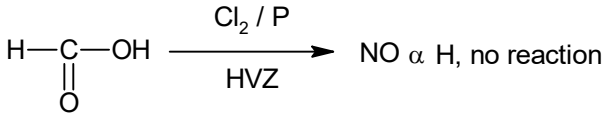
$v_0 = 1 \times 10^{-15}$

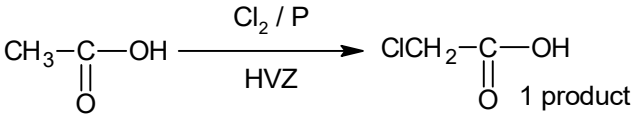
27. 7

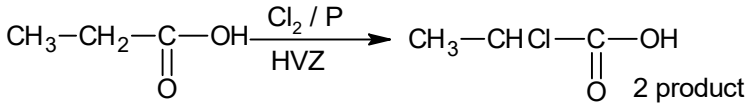
Sol. $P(CH_3)_3(CF_3)_2$, $[PtCl_4]^{2-}$, $XeF_4$, $I_2Cl_6$, $SF_6$, $IF_7$, $[Pt(NH_3)_6]^{+4}$

28. 3

Sol.

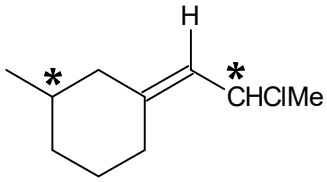



29. 4

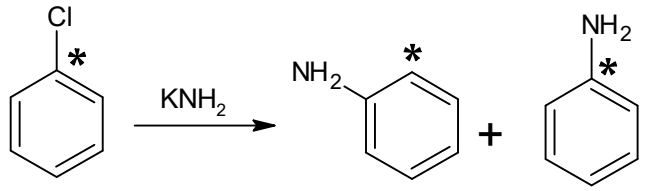
Sol.




$2^2 = 4$

30. 2

Sol.




## SECTION – C

31. 00005.55

Sol. $R-NH_2 + H-OH \rightleftharpoons R-\overset{\oplus}{N}H_3 + \overline{O}H$

$t = 0 \quad C$

$t = eq \quad C - C\alpha \qquad C\alpha \qquad C\alpha$

$$K_b = \frac{C\alpha \times C\alpha}{C - C\alpha} = \frac{C\alpha^2}{1-\alpha}$$

$\because \overline{O}H = C\alpha$

$10^{-3} = 10^{-2}\alpha$

$\alpha = 10^{-1}$

$$\therefore K_b = \frac{10^{-2} \times \left(10^{-1}\right)^2}{1-10^{-1}}$$

$$= \frac{10^{-4}}{0.9}$$

$= \frac{10}{9} \times 10^{-4}$

$x \times 10^{-4} = 1.11 \times 10^{-4}$

$x = 1.11$

$5x = 5.55$

32. 00001.00

Sol. Fact based.

33. 00001.70

Sol. $HA_{aq} \rightleftharpoons H^+_{aq} + A^-_{aq}$

$C - C\alpha \quad C\alpha \quad C\alpha$

$i = 1 + \alpha$

$\Delta T = i \times k_f \times m$

$0.0223 = (1 + \alpha) \times 0.186 \times 0.1$

$1 + \alpha = \frac{0.0223}{0.0186}$

$\alpha = 1.1989 - 1$

$= 0.1989$

$= 0.2$

$pH = -\log C\alpha$

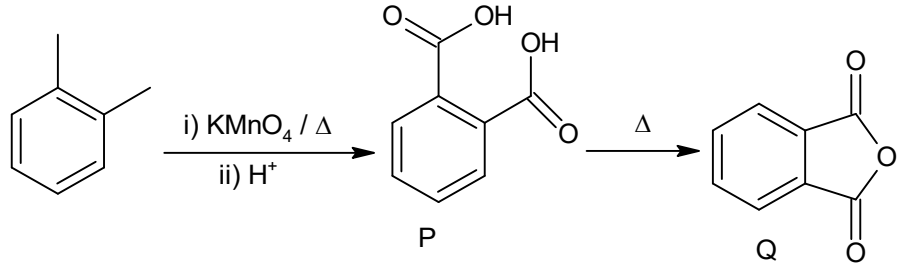
$= -\log 0.1 \times 0.2$

$= -\log 2 \times 10^{-2}$

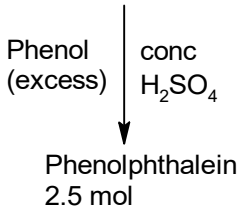
= 1.70

34. 00795.00

Sol.




10 mol 5 mol 2.5 mol



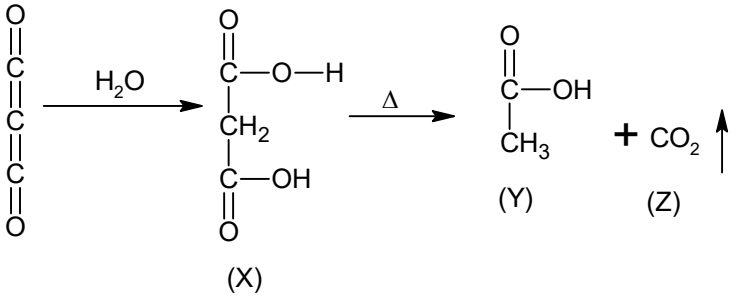

Phenolphthalein
2.5 mol

$W = 2.5 \times 318$

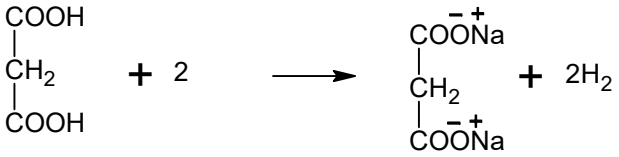
$= 795.00$ g

35. 08800.00

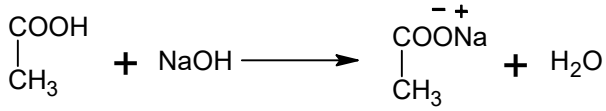
Sol.



(X) (Y) (Z)



$1.1 \times 10^3$ $2 \times 1.1 \times 10^3$



$2.2 \times 10^3$ $2.2 \times 10^3$

$CO_2 + 2 \longrightarrow Na_2CO_3 + H_2O$

$2.2 \times 10^3$ $4.4 \times 10^3$

No. of gram equivalent of $OH^- = 8.8 \times 10^3 = 8800$

36. 09400.00

Sol. (i) $N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$

$n = \frac{310.8 \times 10^3}{28}$ $n = \frac{22.2 \times 10^3}{2}$

$= 11.1 \times 10^3$ $= 11.1 \times 10^3$

$= 11.1 \times 10^3 - \frac{11.1 \times 10^3}{3}$ $= 11.1 \times 10^3 - 11.1 \times 10^3$

$\frac{2}{3} \times 11.1 \times 10^3$

(ii) $CaCO_3 \xrightarrow{\Delta} CO_{2(g)} + CaO_{(s)}$

$n = \frac{100 \times 10^3}{100}$

0 $10^3$

(iii) $NH_4NO_3 \xrightarrow{\Delta} H_2O_{(\ell)} + N_2O_{(g)}$

$n = \frac{80 \times 10^3}{80}$ $10^3$

Total mole of gas $= \frac{2}{3} \times 11.1 \times 10^3 + 10^3 + 10^3 = 9.4 \times 10^3 = 9400$

# Mathematics

## PART – III

### SECTION – A

37. AD

Sol. $y^3 - 3x^2y + x^3 = 0$

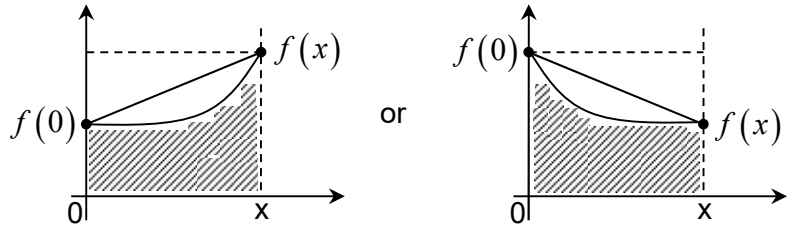
$$\Rightarrow \frac{dy}{dx} = \frac{-\left(-6xy + 3x^2\right)}{\left(3y^2 - 3x^2\right)} = \frac{-3\left(-2xy + x^2\right)}{3\left(y^2 - x^2\right)} = \frac{y}{x}$$

$$\Rightarrow \int\left(\frac{dy}{dx}\right)dx \text{ i.e., } \int\frac{y}{x}dx = y + c$$

$$= f\left(x\right) + c \text{ or } xf'\left(x\right) + c$$

38. BD

Sol.




$$\therefore f'\left(x\right) > 0 \Rightarrow f''\left(x\right) > 0 \text{ or } \therefore f'\left(x\right) < 0 \Rightarrow f''\left(x\right) > 0$$

39. ABCD

Sol. $\left|2A^2 - 2A\right| = 144$

$$2^2\left|A\right|\left|A - I\right| = 144$$

$$\Rightarrow \alpha^2\left(\alpha - 1\right)^2 = 36$$

$$\alpha\left(\alpha - 1\right) = 6 \text{ or } -6$$

$$\alpha^2 - \alpha - 6 = 0 \text{ or } \alpha^2 - \alpha + 6 = 0$$

$$\left(\alpha - 3\right)\left(\alpha + 2\right) = 0 \text{ or } \alpha = \frac{1 \pm \sqrt{1 - 24}}{2 \times 1}$$

$$\alpha = 3 \text{ or } -2 \qquad = \frac{1}{2} \pm i\frac{\sqrt{23}}{2}$$

$$\therefore \text{tr}\left(A\right) = 2\alpha$$

= 6 or –4 or 1 (Real part)

$$\left|A\right| = \alpha^2 = 9 \text{ or } 4$$

40. BC

Sol. If $f\left(a\right) \geq f\left(a - h\right)$ and $f\left(a\right) \geq f\left(a + h\right)$ for $h > 0$ and $h \to 0$, then $x = a$ is a point of maxima.

41. ABC

Sol. $\because$ $f(x) - x^2 = (x-1)^2(x-3)$

42. ABC

Sol. $\because\ f(x)=\dfrac{x}{1+x^2}$ for $x\neq 0$

## SECTION – B

43. 3

Sol. $\because f'(x)=b^2+(a-1)b+2-\left(\sin^2 x+\cos^4 x\right)\geq 0$

$\left(f'(x)\right)_{min}=b^2+(a-1)b+1\geq 0$

$\therefore (a-1)^2-4<0\Rightarrow a\in(-1,\ 3)$

$a=0$ or 1 or 2

$0+1+2=3$

44. 4

Sol. $I=\displaystyle\int_1^2\frac{x^3-3x^2}{\ln(x-1)}$ and $J=\displaystyle\int_2^3\frac{3x-5}{\ln(x-2)}$

$\Rightarrow J=\displaystyle\int_1^2\frac{3x-2}{\ln(x-1)}$

$J+I=\displaystyle\int_1^2\frac{x^3-3x^2+3x-2}{\ln(x-1)}=\int_1^2\frac{(x-1)^3-1}{\ln(x-1)}dx$

$=\displaystyle\int_1^2\frac{(x-1)^3-1}{\ln(x-1)}dx=\int_0^1\frac{x^3-1}{\ln x}dx=\ln 4$

$\therefore e^{\ln 4}=4$

45. 2

Sol. $\overrightarrow{OA}=a\hat{i}+b\hat{j}+c\hat{k}$ and $a^2+b^2+c^2=4$

$\Rightarrow AB=\left(25-24\cos^2\dfrac{\theta}{2}\right)$

$\therefore |QD|=\dfrac{2}{5}\cos\dfrac{\theta}{2}$

$\Rightarrow 25|QD|=2$

46. 4

Sol. $\dfrac{3x^2+1}{\sqrt{x^4+x^2}}=\dfrac{3x^2+1}{x\sqrt{x^2+1}}=\dfrac{x^2+1+2x^2}{x\sqrt{x^2+1}}$

$=\dfrac{\sqrt{x^2+1}}{x}+\dfrac{2x}{\sqrt{x^2+1}}\geq 2\sqrt{2}$

$f(x)=2\sqrt{2}$

$\therefore x^2+1=2x^2$

$x^2=1$

$x=\pm 1 \qquad x=1$

$\int_0^{\sqrt{2}} \frac{\left(3x^2+1\right)dx}{\sqrt{x^4+x^2}} \geq 2\sqrt{2}\times\sqrt{2}=4$

47. 3

Sol. $\because \left|\text{adj}(A)\right| = |A|^{n-1}$

$\Rightarrow \left|\text{adj}\left(\text{adj}(A)\right)\right| = \left|\text{adj}(A)\right|^{n-1} = |A|^{(n-1)^2}$

$\Rightarrow \left|\text{adj}\left(\text{adj}\left(\text{adj}(A)\right)\right)\right| = |A|^{(n-1)^3} \Rightarrow (n-1)^2 = 4$

$n-1=2$

$n=3$

48. 2

Sol. Given $a_{11}=b_{11},\ a_{22}=b_{22},\ a_{12}=-b_{12},\ a_{21}=-b_{21}$

and $AB=BA=I$

$\Rightarrow A=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{11} \end{pmatrix}$ and $|A|=1$

$\therefore a_{11}^2-a_{12}a_{21}=1$

$a_{12}a_{21}=a_{11}^2-1$

$\Rightarrow a_{12}$ or $a_{21}=(a_{11}-1)$ or $(a_{11}+1)$

$\Rightarrow |a_{12}-a_{21}|=2$

## SECTION – C

49. 00002.00

Sol. $\because P_nQ_{n-1}=2\cos 2^{n-1}\theta$

$\therefore P_1Q_0\times P_2Q_1\times.....\times P_nQ_{n-1}=1$

$\frac{2^n\sin\theta\cos\theta\cos 2\theta\cos 2^2\theta.....\cos 2^{n-1}\theta}{\sin\theta}=1$

$\frac{\sin 2^n\theta}{\sin\theta}=1$

$\sin 2^n\theta=\sin\theta$

$2^n\theta=\pi-\theta$

$\Rightarrow \left(2^n+1\right)\theta=\pi$

$\theta=\frac{\pi}{2^n+1}\Rightarrow\frac{\pi}{\theta}=2^n+1$

$\frac{\pi}{\theta.2^{n-1}}=2+\frac{1}{2^{n-1}}$

50. 00001.00

Sol. $\because \begin{vmatrix} x & \sin x & 1+\sin x \\ 1 & \cos x & \cos x \\ 0 & \sin x & \sin x \end{vmatrix} = 0$

$\Rightarrow x = (2n+1)\frac{\pi}{2}$

51. 00000.00

Sol. $\left[f(x) \quad g(x) \quad h(x)\right] = \begin{vmatrix} \cos x & \sin x & \cos 2x \\ \tan x & \sin 3x & \cos 4x \\ \cos 3x & \sin 5x & \cos 6x \end{vmatrix}$

$\therefore \int_0^{\pi} [f, \quad g, \quad h](x)\,dx = \int_0^{\pi} [f, \quad g, \quad h](\pi - x)\,dx$

$\Rightarrow \int_0^{\pi} [f, \quad g, \quad h]\,dx = -\int_0^{\pi} [f, \quad g, \quad h]\,dx \qquad (\because C_1 \to -C_1)$

$\Rightarrow \int_0^{\pi} [f, \quad g, \quad h]\,dx = 0$

52. 00080.00

Sol. $\vec{a}_1 = (a,\ b),\ \vec{a}_2 = \left(a-2,\ b-\sqrt{5}\right),\ \vec{a}_3 = \left(a+2,\ b-\sqrt{5}\right)$

Area of $\Delta$, whose vertices are $\vec{a}_1$, $\vec{a}_2$ and $\vec{a}_3$, is

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}$$

$$\Rightarrow \Delta^2 = \frac{1}{4}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}$$

and $\left|\vec{a}_1 - \vec{a}_2\right| = 3,\ \left|\vec{a}_1 - \vec{a}_3\right| = 3,\ \left|\vec{a}_1 - \vec{a}_3\right| = 4$

$\Rightarrow \Delta = \sqrt{5 \times 2 \times 2 \times 1} = 2\sqrt{5}$

$$\left(2\sqrt{5}\right)^2 = \frac{1}{4}\begin{vmatrix} 1+\vec{a}_1.\vec{a}_1 & 1+\vec{a}_1.\vec{a}_2 & 1+\vec{a}_1.\vec{a}_3 \\ 1+\vec{a}_2.\vec{a}_1 & 1+\vec{a}_2.\vec{a}_2 & 1+\vec{a}_2.\vec{a}_3 \\ 1+\vec{a}_3.\vec{a}_1 & 1+\vec{a}_3.\vec{a}_2 & 1+\vec{a}_3.\vec{a}_3 \end{vmatrix}$$

$\therefore$ Required answer = 4 x 5 x 4 = 80

53. 00003.14

Sol. $\because \left|f(n)\right| \le |0| \Rightarrow f(n) = 0$

& $\left|f(x) - f(n)\right| \le \left|\sin \pi x\right|$

$\Rightarrow \left|f'(n)\right| \le \left|\lim_{x \to n} \cos \pi x \times \pi\right|$

$\Rightarrow \left|f'(n)\right| \le \pi$

54. 00008.00

Sol. $\because \quad \beta = \dfrac{f(4) + f(12)}{2}$

$= \dfrac{4 + 12}{2} = 8$